

Mathematics for Engineers II

SEM II A-C:2016/2017 DURATION: 2 Hours

SECTION A: /20 Marks (compulsory)

1. (a) Show that the Wronskian of any two functions (where one function is a constant multiple of the other) is zero. (2 Marks)
- (b) Reduce the system of the following linear equations into a single high linear equation. $\begin{cases} \frac{dx}{dt} = x + y \\ \frac{dy}{dt} = x - y \end{cases}$ (4 Mark)
- (c) Solve $y' - 5y = e^{5x}$; $y(0) = 0$ by using the Laplace Transform. (4 Marks)
- (d) (i) Find all solutions of $y'' + 2y' + 6y = 0$. (2 Marks)
- (ii) Compute the Laplace Transform of $3t^2 + 5 \cos(4t)$. (2 Mark)
- (e) Determine the Curl of the vector valued function $\vec{F} = x^2y \vec{i} + xyz \vec{j} - x^2y^2 \vec{k}$. Is $\vec{F}$ irrotational? (3 Marks)
- (f) Change the order of integration in

$$\mathcal{I} = \int_0^{4b} \int_{x^2(4b)^{-1}}^{2(bx)^{1/2}} dy dx$$

for all reals b . (3 Marks)

SECTION B: /30 Marks (choose any two questions)

2. (a) State the Divergence Theorem and it to evaluate

$$\int \int_S \vec{F} \cdot d\vec{S}$$

where $\vec{F} = \sin(\pi x) \vec{i} + zy^3 \vec{j} + (z^2 + 4x) \vec{k}$ and S is the surface of the box with $-1 \leq x \leq 2$, $0 \leq y \leq 1$ and $1 \leq z \leq 4$. Notice that all sides of the box are included in S . (8 Marks)

- (b) The population of a certain country is known to increase at a rate proportional to the number of people, N , presently living in the country. If after two year the population has doubled, and after three years the population is 20000, estimate the number of people initially living in the country. (7 Marks)

3. (a) Find the solution of $\frac{dy}{dx}(x) = 3y(x) - 2$, $y(1) = 1$ (4 Marks)

(b) Solve the differential equation $y'' - 2y' + y = e^x/x$ (5 Marks)

(c) Evaluate $\int \int_{\mathbb{R}} xy(x+y) dx dy$ over the region bounded by the curves $y = x^2$ and $y = x$. (6 Marks)

4. (a) Use the matrix method to find the solution to the 2×2 differential system $x' = Ax$,

$$A = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix} \text{ subject to } x(0) = \begin{bmatrix} 1 \\ 5 \end{bmatrix}. \text{ (5 Marks)}$$

(b) Use Laplace Transform to solve the Initial Value Problem $y'' + 3y' + 2y = e^{-t}$ subject to $y'(0) = y(0) = 0$ (5 Marks)

(c) By changing into polar coordinates, evaluate the following integral:

$$\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy$$

(5 Marks)